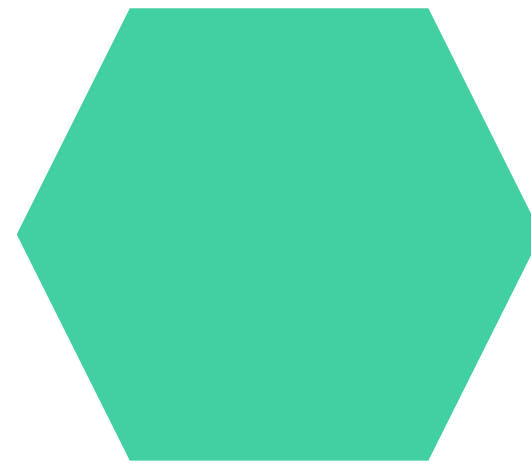
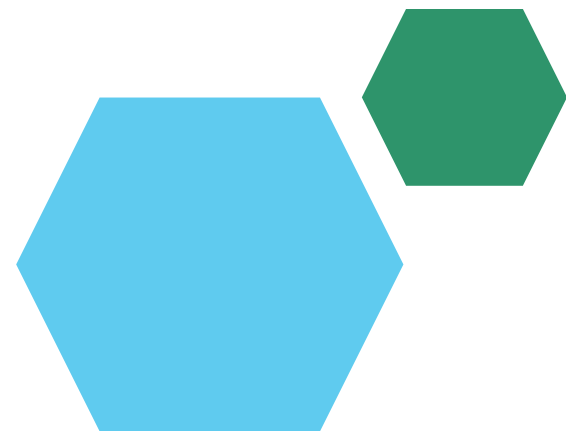




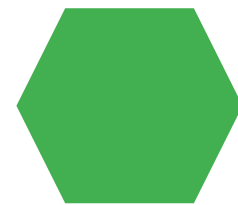
Project Title - Seasonal Agriculture Performance Analysis

A Data-Driven Look at Indian Farming Across Kharif, Rabi and Zaid

Student Name : Jhalak Jain

College Name : Shri Govindram Seksaria Institute of Technology and Science

AICTE STU ID : STU6a156af34a49b1779788531



PROBLEM STATEMENT

- Agricultural activities are influenced by seasonal variations in environmental conditions, farming practices, resource availability and market conditions. As a result, agricultural performance may differ from one season to another.

However, raw agricultural data does not clearly explain how agricultural performance changes across seasons or what patterns can be observed in different seasonal conditions.

The problem is to analyze the given agricultural dataset and investigate seasonal differences in agricultural performance by identifying meaningful patterns, trends, relationships and variations within the available data.



Project Description

This project analyzes 4,000 agricultural records from farms across multiple Indian states.

The dataset includes:

- Environmental factors: rainfall, temperature, soil moisture
- Farming practices: irrigation method, fertilizer, pesticides
- Economic outcomes: profit, revenue, cost
- Resource usage: water consumption and water efficiency

The analysis focuses on one practical question:

Which seasonal combinations of crop, irrigation and farm conditions offer the best balance of profitability and water efficiency?

Tools used: Python with Pandas, NumPy, Matplotlib and Seaborn.

Key techniques applied:

- Median-based comparisons (to handle skewed data)
- IQR method for outlier detection
- Correlation analysis
- A custom "opportunity score" combining profit and water efficiency

WHO ARE THE END USERS?

This analysis is useful for:

1. Farmers

- Know which crops to plant in which season
- Understand which irrigation methods actually work

2. Agricultural Planners

- Identify which crop-season combos are consistently losing money
- Focus support where it's needed most

3. Policy Makers

- Design subsidies for water-efficient practices
- Support vulnerable farming seasons

4. Agribusiness Companies

- Plan supply chains based on seasonal performance patterns

5. Irrigation Departments

- Promote Drip irrigation where it gives the best returns



Technology Used

Python with following tools:

Python 3.13

Jupyter Notebook (for interactive analysis)

Libraries used:

- Pandas → Loading, cleaning, and manipulating data
- NumPy → Working with numbers and arrays
- Matplotlib → Creating charts
- Seaborn → Making charts look professional

Techniques applied:

- Data cleaning (handling missing values)
- IQR method for outlier detection
- Median-based statistics (better for skewed data)
- GroupBy analysis (comparing groups like seasons and crops)
- Correlation analysis (finding relationships)
- Data visualization (bar charts, scatter plots, heatmaps)

Development Environment: VS Code



RESULTS

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Release Notes: 1.136.2Seasonal_Agriculture_Performance_Analysis_Submission_Ready.ipynb

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6. Seasonal comparison

```
data["Profitable"] = data["Profit_INR"] > 0

season_summary = data.groupby("Season").agg(
    Farms=("Farm_ID", "count"),
    Mean_Yield=("Yield_Tonnes_Ha", "mean"),
    Median_Yield=("Yield_Tonnes_Ha", "median"),
    Mean_Profit=("Profit_INR", "mean"),
    Median_Profit=("Profit_INR", "median"),
    Mean_Water=("Water_Used_m3", "mean"),
    Mean_Water_Efficiency=("Water_Efficiency_t_per_1000m3", "mean"),
    Mean_Disease_Risk=("Disease_Pest_Risk_pct", "mean"),
    Profitable_Farm_Rate=("Profitable", "mean")
).reset_index()

season_summary["Profitable_Farm_Rate"] *= 100
season_summary.round(2)
```

[12] ✓ 0.0s

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Release Notes: 1.136.2Seasonal_Agriculture_Performance_Analysis_Submission_Ready.ipynb

Seasonal_Agriculture_Performance_Analysis_Submission_Ready.ipynb>M↓Seasonal Agriculture Performance Analysis>M↓Why I chose this angle>M↓Analytical questions># Basic setupJupyter Notebook

Generate+ Code+ MarkdownRun AllRestartClear All OutputsJupyter VariablesOutlinePython 3.14.6

```
.reset_index()

season_summary["Profitable_Farm_Rate"] *= 100
season_summary.round(2)
```

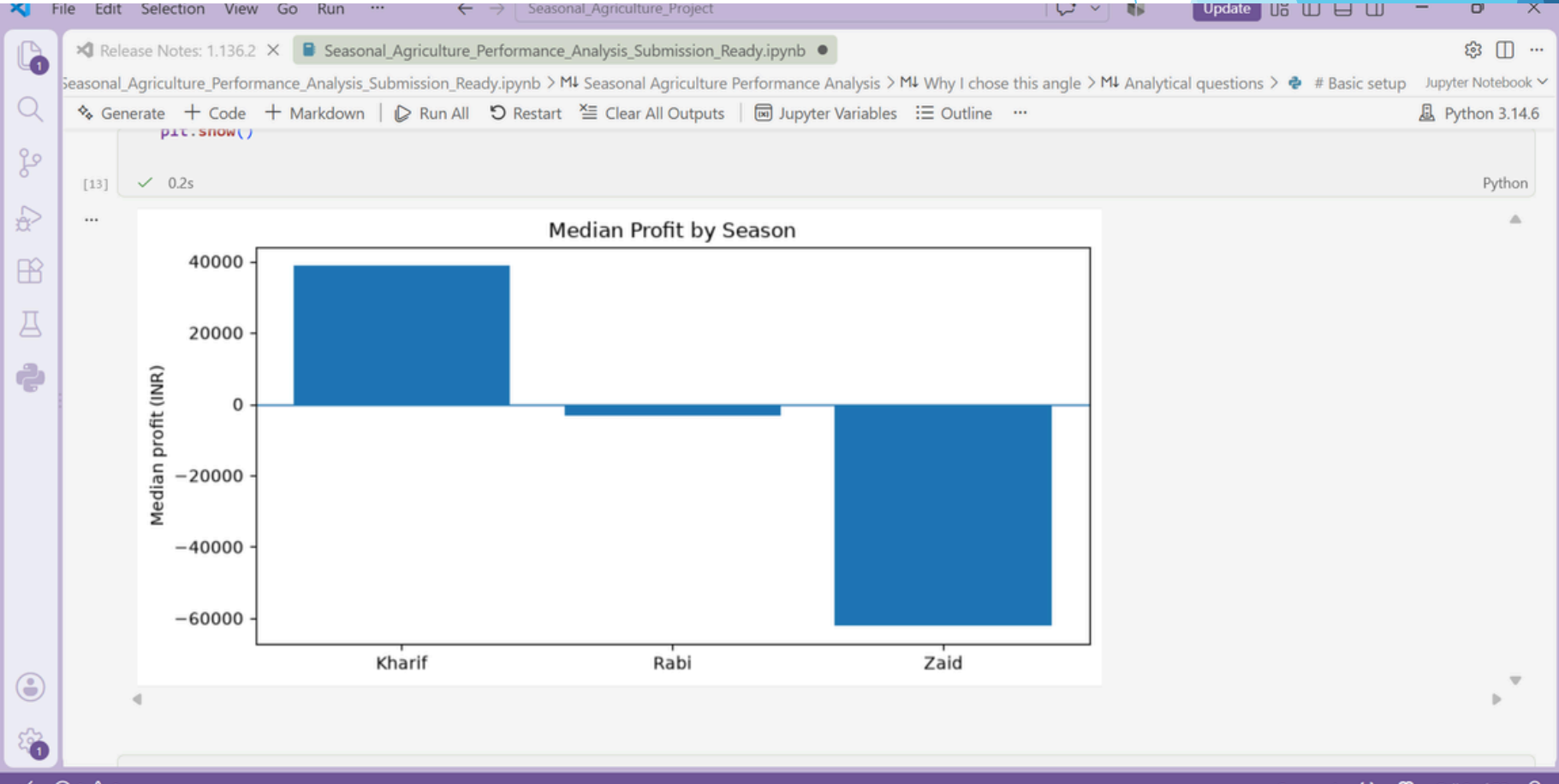
[12] ✓ 0.0s

	Season	Farms	Mean_Yield	Median_Yield	Mean_Profit	Median_Profit	Mean_Water	Mean_Water_Efficiency	Mean_Disease_Risk	Profitable_Farm_Rate
0	Kharif	1779	5.64	1.95	178,914.65	38,808.00	6,102.20	5.89	54.47	57.79
1	Rabi	1627	5.08	1.66	87,689.47	-3,187.00	5,846.99	5.19	40.48	48.86
2	Zaid	594	4.67	1.45	-24,804.82	-62,144.00	6,419.89	4.41	38.22	35.52

```
# Seasonal profit comparison
fig, ax = plt.subplots(figsize=(8, 4))
ax.bar(season_summary["Season"], season_summary["Median_Profit"])
ax.axhline(0, linewidth=1)
ax.set_title("Median Profit by Season")
ax.set_ylabel("Median profit (INR)")
plt.tight_layout()
plt.show()
```

[13] ✓ 0.2s

Median Profit by Season

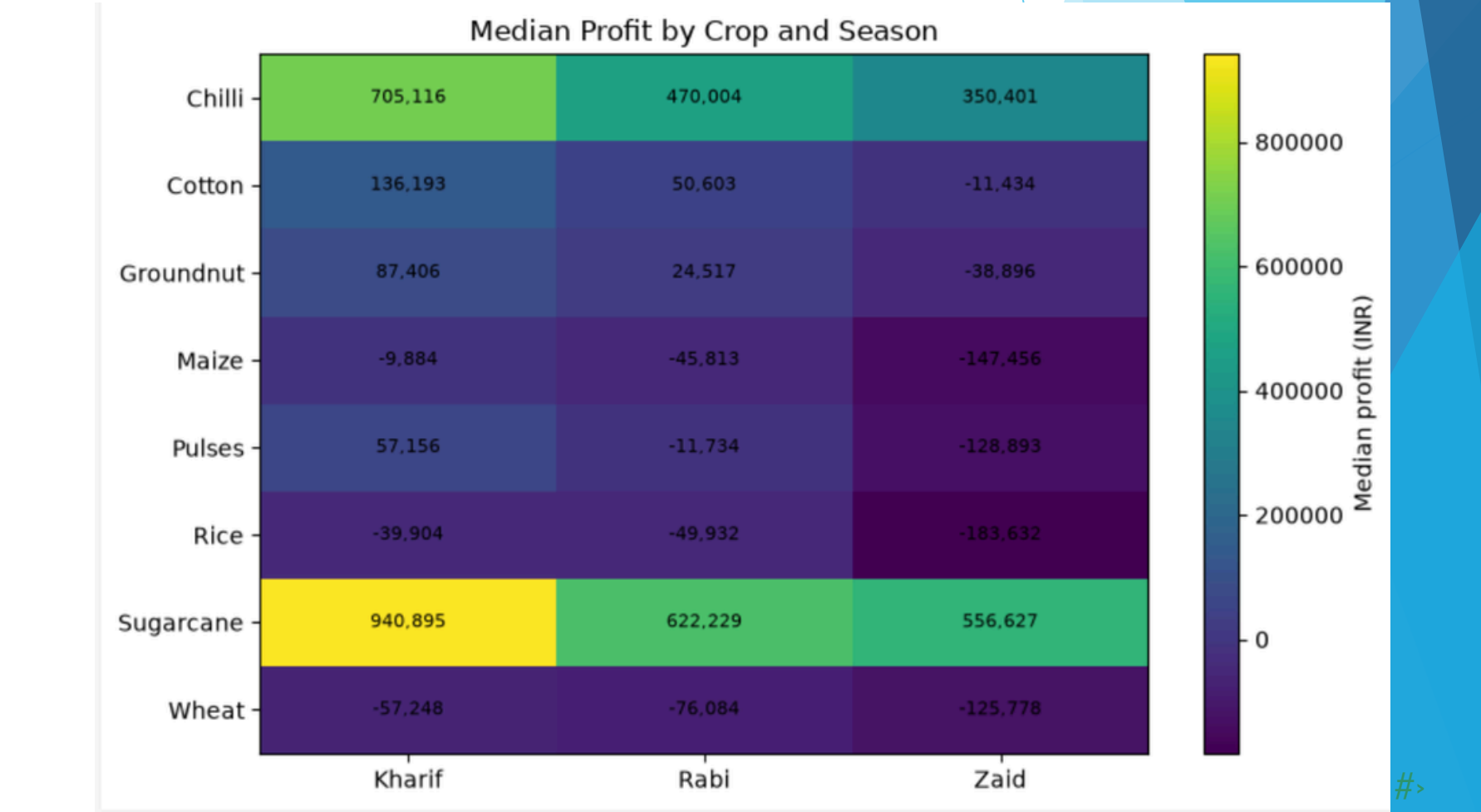


RESULTS

7. Bivariate analysis: crop × season

```
crop_season = data.groupby(["Crop", "Season"]).agg(  
    Farms=("Farm_ID", "count"),  
    Median_Yield=("Yield_Tonnes_Ha", "median"),  
    Median_Profit=("Profit_INR", "median"),  
    Mean_Water_Efficiency=("Water_Efficiency_t_per_1000m3", "mean"),  
    Profitable_Rate=("Profitable", "mean")  
)  
.reset_index()  
  
crop_season["Profitable_Rate"] *= 100  
crop_season.sort_values("Median_Profit", ascending=False).head(15).round(2)
```

```
# Heatmap-like pivot using matplotlib only  
pivot = crop_season.pivot(index="Crop", columns="Season", values="Median_Profit")  
  
fig, ax = plt.subplots(figsize=(8, 5))  
im = ax.imshow(pivot, aspect="auto")  
ax.set_xticks(range(len(pivot.columns)))  
ax.set_xticklabels(pivot.columns)  
ax.set_yticks(range(len(pivot.index)))  
ax.set_yticklabels(pivot.index)  
ax.set_title("Median Profit by Crop and Season")  
fig.colorbar(im, ax=ax, label="Median profit (INR)")  
  
for i in range(pivot.shape[0]):  
    for j in range(pivot.shape[1]):  
        value = pivot.iloc[i, j]  
        ax.text(j, i, f"{value:,.0f}", ha="center", va="center", fontsize=8)  
  
plt.tight_layout()  
plt.show()
```

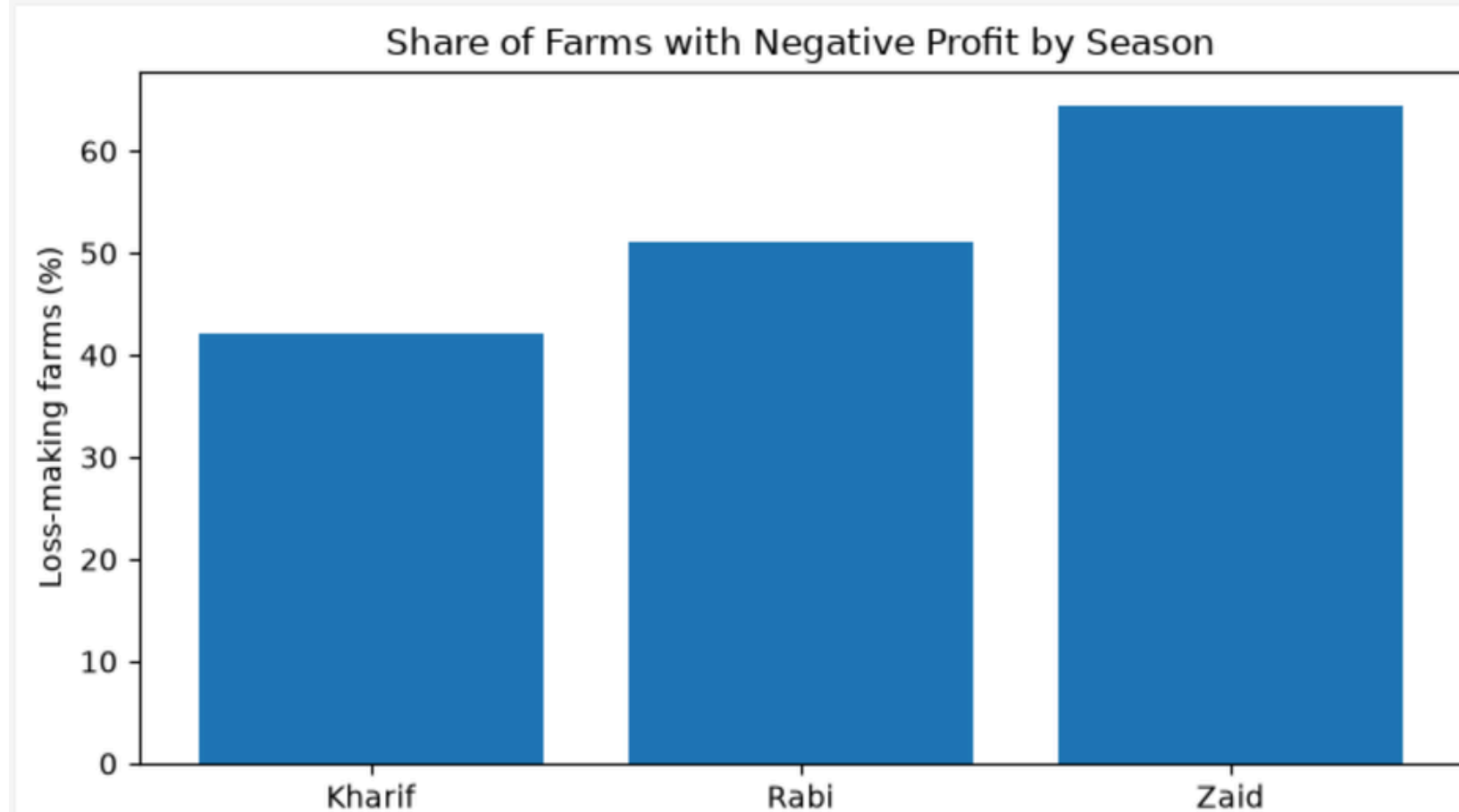


	Crop	Season	Farms	Median_Yield	Median_Profit	Mean_Water_Efficiency	Profitable_Rate
18	Sugarcane	Kharif	125	56.65	940,895.00	36.00	92.00
0	Chilli	Kharif	185	1.78	705,116.00	3.26	85.41
19	Sugarcane	Rabi	129	45.46	622,229.00	27.92	85.27
20	Sugarcane	Zaid	51	38.78	556,627.00	23.39	88.24
1	Chilli	Rabi	163	1.56	470,004.00	2.89	78.53
2	Chilli	Zaid	64	1.22	350,401.00	2.20	81.25
3	Cotton	Kharif	215	1.44	136,193.00	2.13	72.09
6	Groundnut	Kharif	193	1.52	87,406.00	3.58	66.84
12	Pulses	Kharif	221	1.09	57,156.00	3.35	62.90
4	Cotton	Rabi	201	1.20	50,603.00	1.94	66.17
7	Groundnut	Rabi	177	1.29	24,517.00	2.90	56.50
9	Maize	Kharif	234	3.12	-9,883.50	6.05	46.58
5	Cotton	Zaid	92	0.97	-11,434.50	1.49	47.83
13	Pulses	Rabi	213	0.90	-11,734.00	2.70	46.01
8	Groundnut	Zaid	54	1.10	-38,896.50	2.47	40.74

RESULTS

```
# Loss rate by season
season_loss = data.groupby("Season")["Profitable"].apply(lambda s: (1 - s.mean()) * 100)

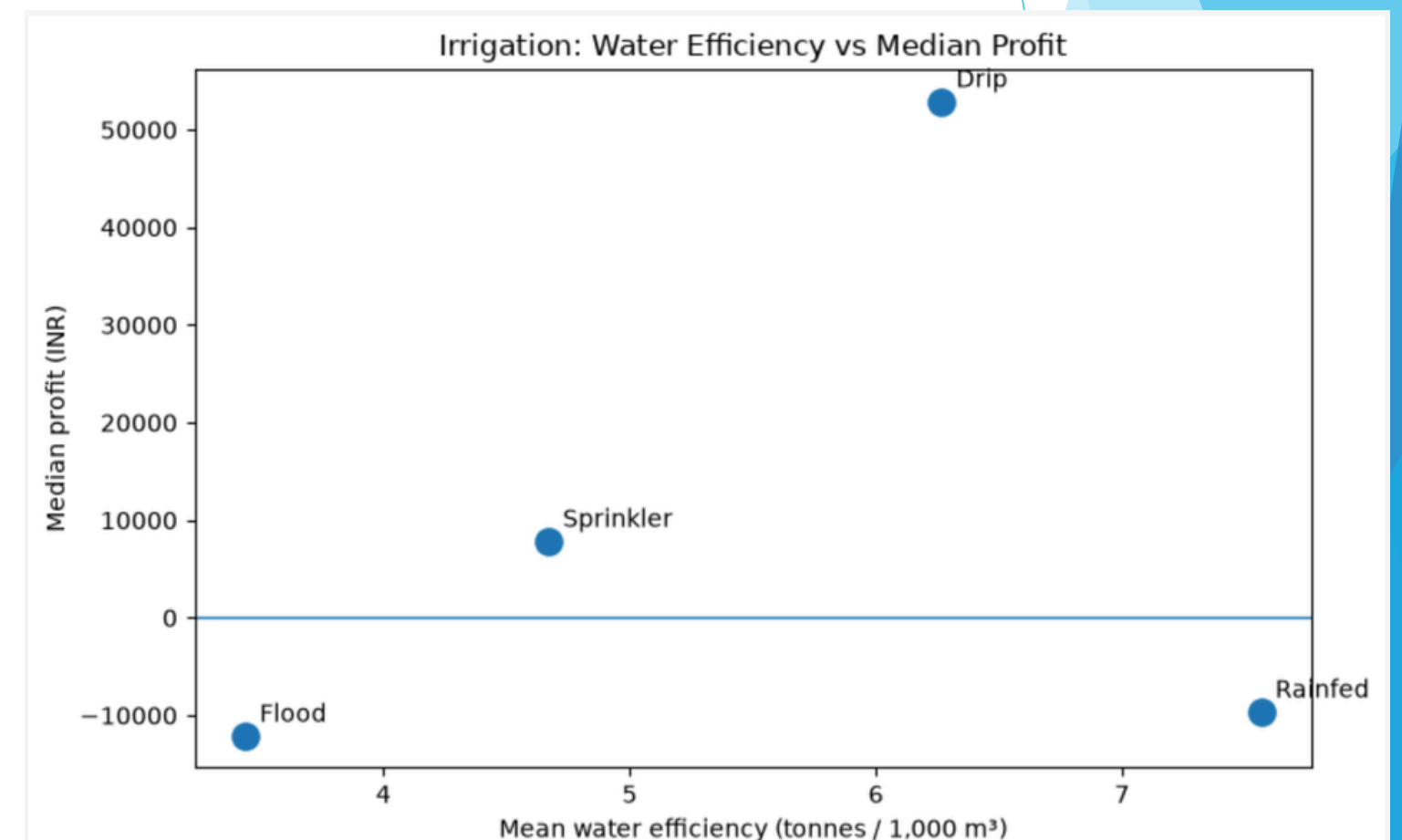
fig, ax = plt.subplots(figsize=(7, 4))
ax.bar(season_loss.index, season_loss.values)
ax.set_title("Share of Farms with Negative Profit by Season")
ax.set_ylabel("Loss-making farms (%)")
plt.tight_layout()
plt.show()
```



```
fig, ax = plt.subplots(figsize=(8, 5))
ax.scatter(
    irrigation_summary["Mean_Water_Efficiency"],
    irrigation_summary["Median_Profit"],
    s=120
)

for _, row in irrigation_summary.iterrows():
    ax.annotate(
        row["Irrigation_Method"],
        (row["Mean_Water_Efficiency"], row["Median_Profit"]),
        xytext=(6, 6),
        textcoords="offset points"
    )

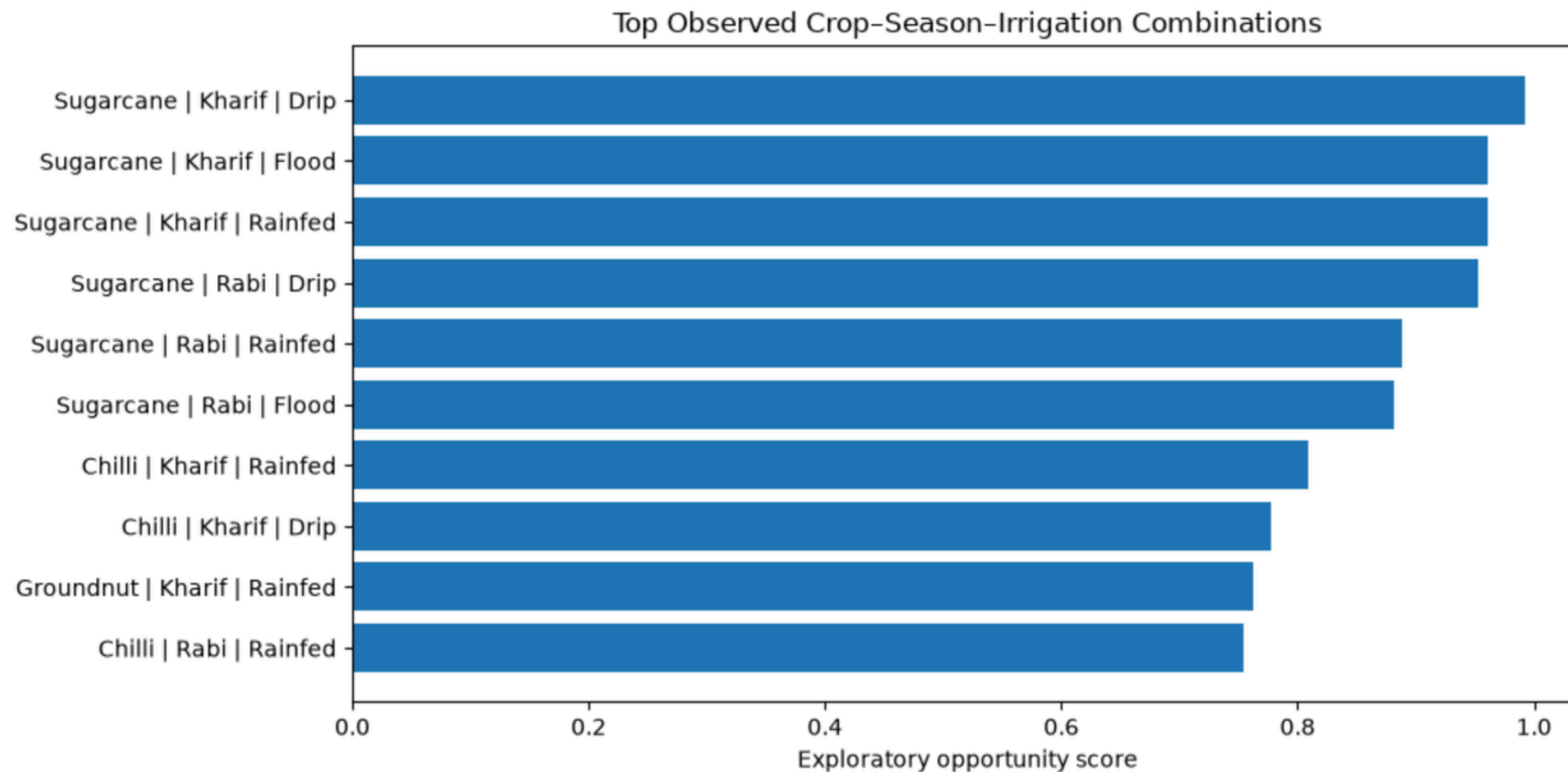
ax.axhline(0, linewidth=1)
ax.set_title("Irrigation: Water Efficiency vs Median Profit")
ax.set_xlabel("Mean water efficiency (tonnes / 1,000 m³)")
ax.set_ylabel("Median profit (INR)")
plt.tight_layout()
plt.show()
```



RESULTS

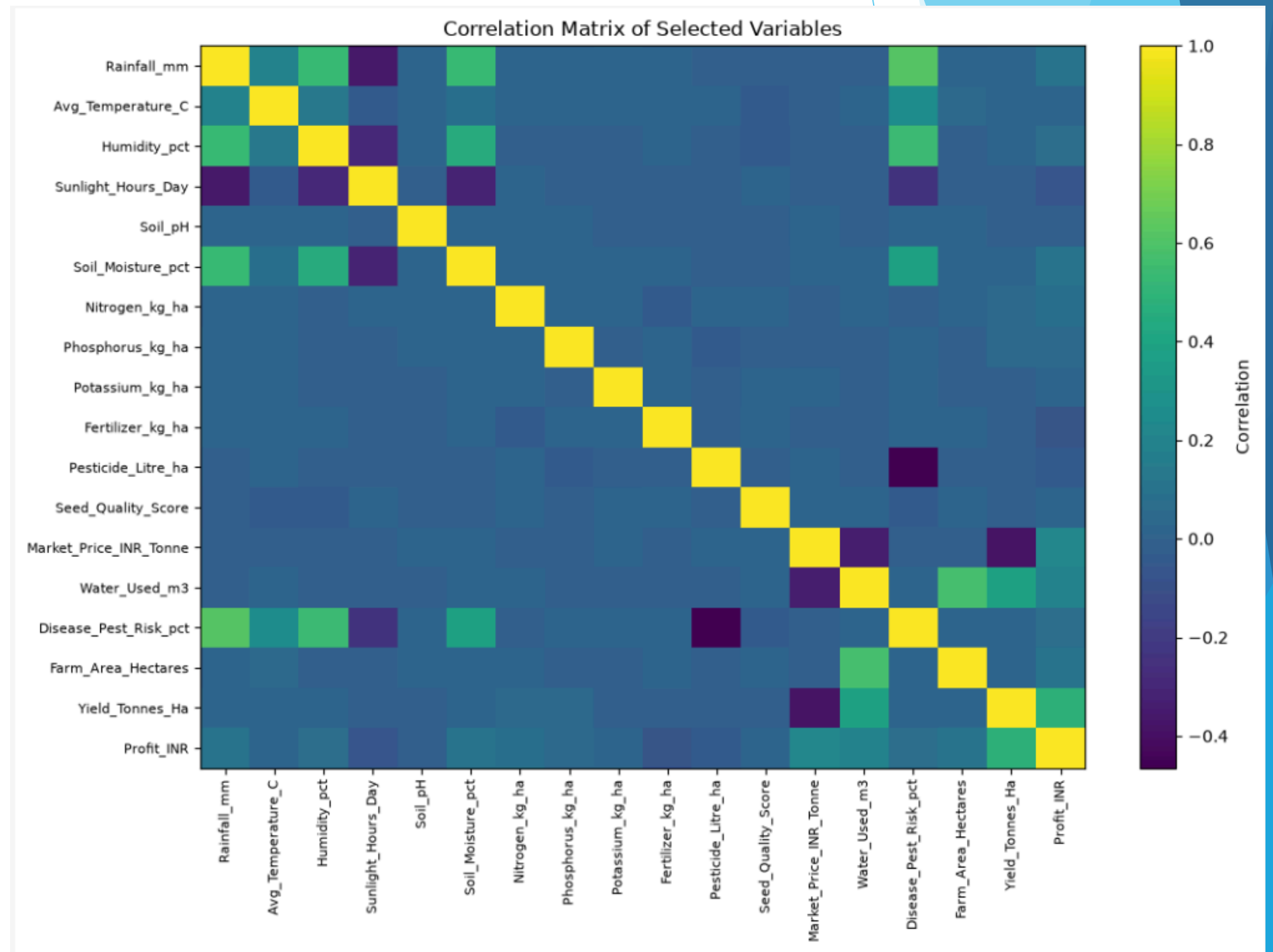
```
# Visualize the highest-ranked combinations
top = opportunity.sort_values("Opportunity_Score", ascending=False).head(10).copy()
top["Label"] = top["Crop"] + " | " + top["Season"] + " | " + top["Irrigation_Method"]

fig, ax = plt.subplots(figsize=(10, 5))
ax.barh(top["Label"][::-1], top["Opportunity_Score"][::-1])
ax.set_title("Top Observed Crop-Season-Irrigation Combinations")
ax.set_xlabel("Exploratory opportunity score")
plt.tight_layout()
plt.show()
```



RESULTS

```
# Correlation heatmap
fig, ax = plt.subplots(figsize=(11, 8))
im = ax.imshow(corr.values, aspect="auto")
ax.set_xticks(range(len(corr.columns)))
ax.set_xticklabels(corr.columns, rotation=90, fontsize=8)
ax.set_yticks(range(len(corr.index)))
ax.set_yticklabels(corr.index, fontsize=8)
ax.set_title("Correlation Matrix of Selected Variables")
fig.colorbar(im, ax=ax, label="Correlation")
plt.tight_layout()
plt.show()
```



Future scope

HOW THIS COULD BE EXTENDED

1. Machine Learning Models
 - Predict yield or profit based on seasonal conditions
2. Regional (State-Level) Analysis
 - Compare performance across different Indian states
3. Climate Change Impact
 - Study how changing weather patterns affect seasons
4. Interactive Dashboard
 - Build a tool for farmers to explore crop-season combinations
5. Environmental Cost Analysis
 - Include water depletion and soil degradation costs
6. Price Forecasting
 - Predict market prices based on seasonal supply patterns

What can be done next:

Build a simple seasonal planning dashboard and test these patterns with real farmers.

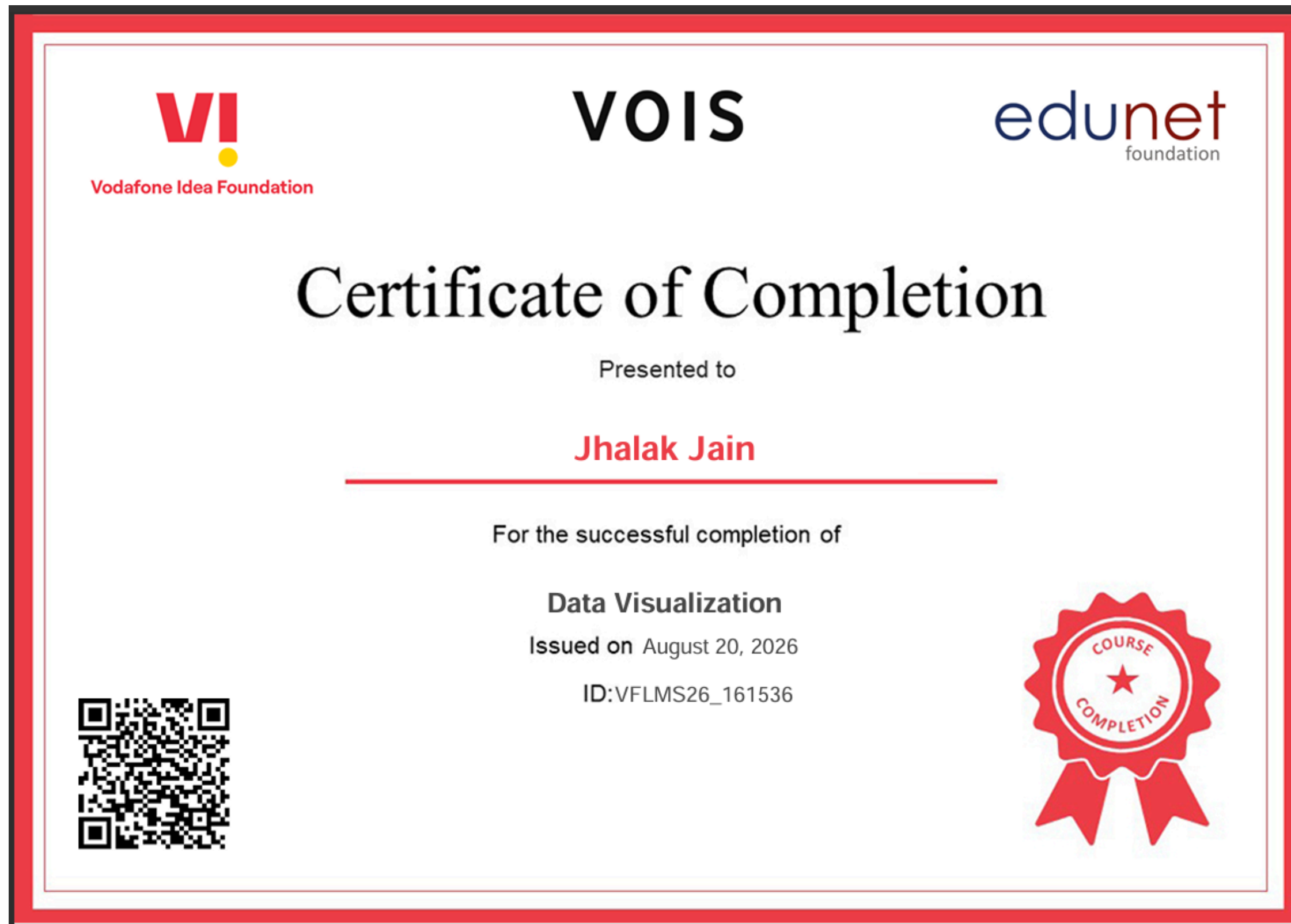
GitHub Link



<https://github.com/jhalak2006-dotcom/Seasonal-Agriculture-Performance-Analysis>



VOIS Course completion certificate - Course name- Data Visualization



Thank you

"This analysis was done as part of
the VOIS Data Analytics Internship
Major Project"

